Python Practice Worksheet

Name: ----------
Class: ----------
Date: ----------

Part 1: Predict the Output (30 Questions)

1. `x, y, z = 10, 20, 30; x, y, z = y, z, x; print(x, y, z)`

   (Hint: Think about how Python evaluates the entire right side of a multiple assignment before binding the variables.)

   a. 10 20 30
   b. 20 30 10
   c. 30 10 20
   d. Error

2. `a = b = [1, 2]; a.append(3); print(b)`

   (Hint: Chained assignments create multiple references to the same object in memory.)

   a. [1, 2]
   b. [3]
   c. [1, 2, 3]
   d. Error

3. `print("A", "B", sep="*", end=""); print("C")`

   a. A*BC
   b. A*B C
   c. A B*C
   d. A*B\nC

4. `import math; print(math.ceil(-4.2), math.floor(-4.2))`

   (Hint: Think about which direction is "up" or "down" on the negative side of the number line.)

   a. -4 -5
   b. -5 -4
   c. -4 -4
   d. -5 -5

5. `print(0 or "" or [] or "Found" or "Lost")`

   (Hint: The 'or' operator returns the first truthy value it encounters.)

   a. 0
   b. Found
   c. Lost
   d. True

6. `print(5 and [1] and 0 and "Done")`

   (Hint: The 'and' operator returns the first falsy value it encounters.)

   a. 5
   b. [1]
   c. 0
   d. Done

7. `print(2 ** 2 ** 3)`

(Hint: Exponentiation associates from right to left.)

a. 64
b. 256
c. 128
d. 512

8. `t = (1, 2); print(id(t) == id(t + (3,)))`

a. True
b. False
c. TypeError
d. ValueError

9. `d = {'x': 10}; print(d.get('y', d.get('x')))`

a. None
b. 10
c. 'x'
d. KeyError

10. `s1 = {1, 2, 3}; s2 = {3, 4}; print(s1.intersection(s2))`

a. {3}
b. {1, 2, 3, 4}
c. {1, 2, 4}
d. {}

11. `print(10 + 10 // 3 * 2)`

a. 13.33
b. 16
c. 13
d. 12.0

12. `L = [1, 2, 3, 4]; L[1:3] = [9]; print(L)`

(Hint: Slices can be replaced with iterables of different lengths.)

a. [1, 9, 4]
b. [1, 9, 3, 4]
c. [1, [9], 4]
d. TypeError

13. `print(int("10" + "20") * 2)`

a. 2040
b. 10201020
c. 60
d. 2040.0

14. `x = [1, 2]; y = [1, 2]; print(x == y, x is y)`

(Hint: == checks value equality, is checks memory address identity.)

a. True True
b. True False
c. False True
d. False False

15. `x = 5; print(f"{x * 2 = }")`

(Hint: This is a special debugging feature in modern Python f-strings.)

a. 10
b. x * 2 = 10
c. 5 * 2 = 10
d. SyntaxError

16. `d = {1: 'A'}; d.update({1: 'B', 2: 'C'}); print(d)`

a. {1: 'A', 2: 'C'}
b. {1: 'A', 1: 'B', 2: 'C'}
c. {1: 'B', 2: 'C'}
d. TypeError

17. `x = 10; print("High" if x > 15 else "Mid" if x > 5 else "Low")`

a. High
b. Mid
c. Low
d. SyntaxError

18. `print(-12 % 5)`

(Hint: In Python, the modulo result always shares the sign of the divisor.)

a. -2
b. 2
c. 3
d. -3

19. `word = "radar"; print(word[0] == word[-1] and word[1:4] == word[1:4][::-1])`

a. True
b. False
c. SyntaxError
d. IndexError

20. `s = {1, 2, 3}; s.discard(4); s.remove(2); print(s)`

(Hint: Understand the difference between discard and remove when an item is missing.)

a. {1, 3}
b. KeyError
c. {1, 2, 3}
d. AttributeError

21. `print(bool([0]), bool(0.0))`

a. False False
b. True True
c. True False
d. False True

22. `import math; print(math.fabs(-7))`

(Hint: math.fabs always returns a specific numeric type.)

a. 7
b. 7.0
c. -7.0
d. TypeError

23. `print(int(True) + int(False))`

a. True
b. 1
c. 0
d. TypeError

24. `a, b = 0, 1; a, b = b, a + b; print(a, b)`

a. 1 1
b. 0 1
c. 1 0
d. 1 2

25. `s = "Internship"; print(s[1:8:2])`

a. n r h
b. n e n h
c. t r s i
d. e s i

26. `t = (1, 2, [3, 4]); t[2].extend([5]); print(t)`

(Hint: A tuple is immutable, but what about the internal state of mutable objects it references?)

a. TypeError
b. (1, 2, [3, 4])
c. (1, 2, [3, 4, 5])
d. AttributeError

27. `print(1 < 2 < 3 == 3 > 1)`

(Hint: Python chains relational operators with a hidden 'and' between each comparison.)

a. True
b. False
c. SyntaxError
d. TypeError

28. `print(True + True * False - True)`

a. True
b. False
c. 1
d. 0

29. `x, y = 5, 10; print((x, y) == (y, x))`

a. True
b. False
c. TypeError
d. SyntaxError

30. `L = [[0]] * 3; L[0][0] = 1; print(L)`

(Hint: Multiplying a list duplicates references, not the actual nested objects.)

a. [[1], [0], [0]]
b. [[1], [1], [1]]
c. [[0], [0], [0]]
d. TypeError

Part 2: Find the Error (20 Questions)

31. `if x == 10\n print(x)`

a. Missing colon (:)
b. IndentationError
c. NameError
d. ValueError

32. `conda create environment myenv python=3.10`

a. python=3.10 should be python==3.10
b. 'environment' is invalid syntax; use -n
c. Missing quotes around environment name
d. No error

33. `True = 5`

a. SyntaxError (cannot assign to keyword)
b. TypeError
c. NameError
d. ValueError

34. `my_set = {[1, 2], 3}`

(Hint: Sets require their elements to meet a specific memory requirement.)

a. TypeError (unhashable type: 'list')
b. SyntaxError
c. ValueError
d. No error

35. `d = {[1]: 'a'}`

a. KeyError
b. TypeError (unhashable type: 'list')
c. SyntaxError
d. No error

36. `t = (1, 2); t[0] = 3`

a. IndexError
b. TypeError (tuple does not support assignment)
c. SyntaxError
d. ValueError

37. `s = "Code"; s[1] = "O"`

a. SyntaxError
b. IndexError
c. TypeError ('str' does not support assignment)
d. No error

38. `print(int("15.5"))`

(Hint: The int() function has strict rules about string literal formats.)

a. TypeError
b. ValueError
c. SyntaxError
d. No error

39. `age = input("Age: "); print(age + 5)`

(Hint: Consider what data type the input function always returns by default.)

a. NameError
b. ValueError
c. TypeError
d. SyntaxError

40. `print(ceil(5.5))`

a. NameError (missing 'math.' prefix or import)
b. TypeError
c. ValueError
d. No error

41. `x = 5; elif x > 3: print(x)`

a. IndentationError
b. SyntaxError (elif must follow an if)
c. NameError
d. TypeError

42. `val = 10; print(f"Result: {value}")`

a. SyntaxError
b. NameError (name 'value' is not defined)
c. ValueError
d. No error

43. `t = (1, 2); t.append(3)`

a. TypeError
b. AttributeError ('tuple' has no attribute 'append')
c. SyntaxError
d. IndexError

44. `print(True and 5 / 0)`

a. ZeroDivisionError
b. TypeError
c. ValueError
d. No error

45. `a, b, c = 10, 20`

a. SyntaxError
b. ValueError (not enough values to unpack)
c. TypeError
d. No error

46. `import math; print(math.fabs("Hello"))`

a. SyntaxError
b. ValueError
c. TypeError (must be real number, not str)
d. No error

47. `s1 = {1}; s2 = {2}; print(s1 + s2)`

(Hint: Sets do not support mathematical-style concatenation operators.)

a. TypeError (unsupported operand for +)
b. ValueError
c. SyntaxError
d. KeyError

48. `t = (1, 2, [3, 4]); t[2] += [5, 6]`

(Hint: The += operator modifies a list in-place, but assigning it back to the tuple index triggers an exception. Which one?)

a. TypeError (but the list is modified in memory)
b. MemoryError
c. AttributeError
d. ValueError

49. `x = 5; print(f"{x:.2f}" + f"{x:.2d}")`

a. ValueError (precision not allowed in integer 'd')
b. TypeError
c. SyntaxError
d. No error

50. `L = [1, 2, 3]; L = L.append(4); print(L[0])`

(Hint: What does the .append() method actually return when it finishes running?)

a. IndexError
b. SyntaxError
c. TypeError ('NoneType' is not subscriptable)
d. ValueError